

Disproof of conjecture 00000000443

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2026-09-12

Verdict: FALSE

We disprove conjecture 00000000443 as stated. There are two *independent* failures:

- (A) the asserted lower bound $\dim L_n \geq 2^{n-1} - 2^{\lceil n/2 \rceil}$ is false for *every* $n \geq 4$ (it first fails at $n = 4$);
- (B) the asserted prime-gap formula “the gap is exactly half of $2^{(n-1)/2}$ ” is false at $n = 3, 7, 11$ (it happens to hold at $n = 5$, and is not even integral at $n = 2$).

Each failure alone falsifies the conjecture. The exact dimensions used below are the standard ones, given by Witt’s formula; the disproof is pure arithmetic and requires no structure theory.

1 The conjecture

English. Definition: The n -th homogeneous dimension of the free Lie algebra $L(V)$ is given by Witt’s formula $\frac{1}{n} \sum_{d|n} \mu(d) k^{n/d}$. Conjecture: For $k = 2$ and all $n \geq 2$, $\dim L_n \geq 2^{n-1} - 2^{\lceil n/2 \rceil}$, and for n prime the gap between this bound and the exact value is exactly half of $2^{(n-1)/2}$.

The statement as filed in `conjectures/00000000443.md` also carries a Chinese translation; it is reproduced in the accompanying `README.md`.

Throughout, L_n denotes the degree- n homogeneous component of the free Lie algebra on two generators ($k = 2$), and μ is the Möbius function.

2 Witt’s formula and the exact dimensions

Theorem 1 (Witt). *For the free Lie algebra on k generators, the homogeneous component of degree $n \geq 1$ has dimension*

$$\dim L_n = \frac{1}{n} \sum_{d|n} \mu(d) k^{n/d}.$$

We use the specialisation $k = 2$. The values for $n = 1, \dots, 12$ are given in Table 1. For example, at $n = 4$ the divisors are 1, 2, 4 with $\mu(1) = 1$, $\mu(2) = -1$, $\mu(4) = 0$, so

$$\dim L_4 = \frac{1}{4} (\mu(1)2^4 + \mu(2)2^2 + \mu(4)2^1) = \frac{1}{4} (16 - 4 + 0) = \frac{12}{4} = 3.$$

n	1	2	3	4	5	6	7	8	9	10	11	12
$\dim L_n$	2	1	2	3	6	9	18	30	56	99	186	335

Table 1: Exact dimensions $\dim L_n$ for $k = 2$, $n = 1, \dots, 12$, from Witt's formula (Theorem 1).

3 The claimed bound

Write

$$B(n) = 2^{n-1} - 2^{\lceil n/2 \rceil},$$

so the conjecture asserts $\dim L_n \geq B(n)$ for all $n \geq 2$. Table 2 compares $B(n)$ with $\dim L_n$.

n	$\dim L_n$	$B(n) = 2^{n-1} - 2^{\lceil n/2 \rceil}$	$\dim L_n \geq B(n)?$
1	2	$2^0 - 2^1 = -1$	yes
2	1	$2^1 - 2^1 = 0$	yes
3	2	$2^2 - 2^2 = 0$	yes
4	3	$2^3 - 2^2 = 4$	no
5	6	$2^4 - 2^3 = 8$	no
6	9	$2^5 - 2^3 = 24$	no
7	18	$2^6 - 2^4 = 48$	no
8	30	$2^7 - 2^4 = 112$	no
9	56	$2^8 - 2^5 = 224$	no
10	99	$2^9 - 2^5 = 480$	no
11	186	$2^{10} - 2^6 = 960$	no
12	335	$2^{11} - 2^6 = 1984$	no

Table 2: The claimed bound $B(n)$ against the exact dimension. The inequality holds only for $n = 2, 3$ and fails for every $4 \leq n \leq 12$. (At $n = 1$ the formula gives a negative value, but the conjecture only ranges over $n \geq 2$.)

4 Failure (A): the inequality is false for every $n \geq 4$

Lemma 2. $\dim L_4 = 3$.

Proof. By Theorem 1 with $k = 2$ and $n = 4$: the positive divisors of 4 are 1, 2, 4, and $\mu(1) = 1$, $\mu(2) = -1$, $\mu(4) = 0$. Hence

$$\dim L_4 = \frac{1}{4}(1 \cdot 2^4 + (-1) \cdot 2^2 + 0 \cdot 2^1) = \frac{1}{4}(16 - 4) = \frac{12}{4} = 3. \quad \square$$

Lemma 3. $2^3 - 2^2 = 4$, i.e. $B(4) = 4$.

Proof. $2^3 = 8$ and $2^2 = 4$, so $2^3 - 2^2 = 8 - 4 = 4$. Since $\lceil 4/2 \rceil = 2$, this is exactly $B(4)$. $\square$

Theorem 4 (failure of the inequality). $\dim L_4 = 3 < 4 = B(4)$. Consequently the conjecture's assertion " $\dim L_n \geq 2^{n-1} - 2^{\lceil n/2 \rceil}$ for all $n \geq 2$ " is false.

Proof. By Lemma 2, $\dim L_4 = 3$. By Lemma 3, $B(4) = 4$. Since $3 < 4$, the inequality fails at $n = 4$. A single counterexample refutes a universal statement. In fact Table 2 shows the inequality fails at every $n \in \{4, 5, \dots, 12\}$. $\square$

Corollary 5. *The conjunction forming conjecture 00000000443 is false.*

Proof. The conjecture asserts a universally quantified inequality; Theorem 4 exhibits $n = 4$ for which it fails. A conjunction containing a false conjunct is false. $\square$

5 Failure (B): the prime-gap formula is false

The conjecture further asserts that for prime n the gap between the bound and the exact value is *exactly* half of $2^{(n-1)/2}$, i.e. (taking the magnitude of the gap)

$$|B(n) - \dim L_n| = \frac{1}{2} \cdot 2^{(n-1)/2} \quad (n \text{ prime}).$$

This is a second, logically independent assertion. Table 3 evaluates both sides for the primes $n = 2, 3, 5, 7, 11$.

n (prime)	$\dim L_n$	$B(n)$	$ B(n) - \dim L_n $	$\frac{1}{2}2^{(n-1)/2}$	claim holds?
2	1	0	1	(not an integer)	ill-defined
3	2	0	2	1	no
5	6	8	2	2	yes (coincidence)
7	18	48	30	4	no
11	186	960	774	16	no

Table 3: The prime-gap assertion. It fails at $n = 3, 7, 11$, holds coincidentally at $n = 5$, and is not even an integer at $n = 2$.

Theorem 6 (failure of the prime-gap formula). *The assertion “for prime n the gap is exactly half of $2^{(n-1)/2}$ ” is false. Explicitly:*

$$n = 3 : \quad |B(3) - \dim L_3| = |0 - 2| = 2 \neq 1 = \frac{1}{2}2^1,$$

$$n = 7 : \quad |B(7) - \dim L_7| = |48 - 18| = 30 \neq 4 = \frac{1}{2}2^3,$$

$$n = 11 : \quad |B(11) - \dim L_{11}| = |960 - 186| = 774 \neq 16 = \frac{1}{2}2^5.$$

Proof. The values $\dim L_3 = 2$, $\dim L_7 = 18$, $\dim L_{11} = 186$ are read from Table 1; the bounds are $B(3) = 0$, $B(7) = 48$, $B(11) = 960$ from Table 2. Each displayed inequality is then integer arithmetic. Since the claimed identity fails at three primes, it is false as a general statement. $\square$

Remark 7. At $n = 5$ we have $|B(5) - \dim L_5| = |8 - 6| = 2 = \frac{1}{2}2^2$, so the formula happens to be correct there; a single coincidence cannot support a claim quantified over all primes, and $n = 3, 7, 11$ already refute it. At $n = 2$ the quantity $\frac{1}{2}2^{(n-1)/2} = \frac{1}{2}\sqrt{2}$ is irrational, while the gap is the integer 1, so the phrase “exactly half of $2^{(n-1)/2}$ ” cannot hold at $n = 2$ under any reading either.

6 The two failures are independent

Failure (A) is a statement about the *inequality* and is witnessed at the composite value $n = 4$ (and all $n \geq 4$ in the tested range). Failure (B) is a statement about the *prime gap* and is witnessed at the primes $n = 3, 7, 11$. Either failure alone makes the conjecture false: the conjecture is the conjunction of (i) the universal inequality and (ii) the prime-gap identity, and a conjunction is false as soon as one conjunct is false. Accordingly, even a reader who doubted one of the two computations would still be forced to reject the conjecture by the other.

7 Reproducing the computation

The numerical content is reproduced by `reproduce.py`, which uses only the Python 3 standard library and implements the Möbius function by hand:

```
python3 reproduce.py
```

It prints Tables 1–3, checks that $\dim L_n < B(n)$ for all $n \in [4, 12]$, checks that the gap claim fails at $n = 3, 7, 11$ (and notes the coincidence at $n = 5$), and exits with status 0 (**PASS**) only if every check succeeds.